

Emergency Supply Chain Game

How to Play

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Task

We are experiencing flash flood that has disrupted the normal water supply, creating an urgent need to manage emergency water distribution effectively.

Federal Stockpile -> Regional Hubs -> State/Local Hubs -> Shelters

Water moves **downstream** toward Shelters. Orders and requests move **upstream** toward the Federal Stockpile.

Use the information available to you to make the best order decision you can.

The Four Tiers

Tier	Main responsibility
Federal Stockpile	Supply water to Regional Hubs
Regional Hubs	Supply water to State/Local Hubs
State/Local Hubs	Supply water to Shelters
Shelters	Meet the people's demand for water

Each person plays one tier. Your order becomes an incoming request for the tier above you.

Your Goal

Each game day, choose how many units of water to order.

You are balancing two competing risks:

- Order too little and unmet requests (demand) become **backorders**.
- Order too much and unused water remains as **inventory**.

Your aim as a team is to ensure Shelters have enough water to meet the people's demand.

Step 1: Join the Game

1. Choose your assigned **Team Number** and **Team Letter**.
2. Select your assigned role.

Step 2: Understand Lag

Orders and shipments do **not** arrive immediately.

A disaster can disrupt deliveries. If that happens, the lag may be **longer than usual**.

Under normal conditions, the lag is 1 day for an order to travel upstream and 1 day for water to travel downstream.

Example: with a 1-day lag, if you order 1 unit of water in Day 1, it takes 1 day for the order information to travel upstream, and another 1 day for the water to travel downstream. Thus the water will arrive on Day 3.

Step 3: Understand Cost

Costs are recorded at the end of every game day.

Cost	How it is created
Inventory cost	\$0.10 for each unit held in inventory at the end of the day
Backlog cost	\$0.20 for each unit in backorder at the end of the day
Shared shortage cost	\$0.50 charged to every team member each day that Shelters have any backlog

Total cost = Inventory cost + Backlog cost + Shared shortage cost

You receive **10% of your remaining budget** as your bonus payment.

Bonus Payment = 10% of (Starting budget - Total cost)

For example, if your remaining budget is \$70 at the end of the game, you receive \$7. The lower your total cost, the more budget you have left and the higher your bonus payment.

Step 4: Submit Your Order

When it is your turn:

1. Enter an order from **0 to 1,000** units in the Daily Order Decision field.
2. Select **SUBMIT**.
3. Confirm that your order is recorded, then wait for the other tiers.

Your submitted order is final for that game day.

Example

Use the **Record Sheet** to review what happened on previous game days.

Day	Demand	Order	Arrived	Shipped	Inventory	Backlog
1	8	4	0	8	0	0
2	4	5	0	0	0	4
3	4	10	3	3	0	5
4	4	15	9	9	0	0

Questions before we start?

Step 5: Complete the Trial

You will now complete a short trial before the full game begins.

Step 6: Play the Game

When you reach the end screen, do not restart the game or close the page. Wait for the facilitator's next instruction.